Dash node TH PCB (hw/dash, rev 2.0): 30-pin ESP32 DevKit = Pi supervisor + haptic knob + CAN, 2DIN opening (D47).

Dash node



File: dash-node.kicad_sch

Power: 3V3 from power module J5 (J20). 3V3_SW (Q10) feeds the CAN transceiver + AS5600, off in sleep.
CAN junction: ME221 (J21) + RJ45 to main (J12 pins 1/2) + Pi HAT stub (J22); no 120 ohm on the dash (mid-bus).
Knob: SimpleFOC Mini off-board on cable J19 (VM from module J2), AS5600 (J24, I2C), push switch (J25).
Car inputs IGN/ACC (J26) through inverting NPN stages; Pi handshake on J23.
TH parts owner-fitted; U11/C20/D20 (SMD) placed by JLCPCB. Module pinouts [E]: check against the boards.

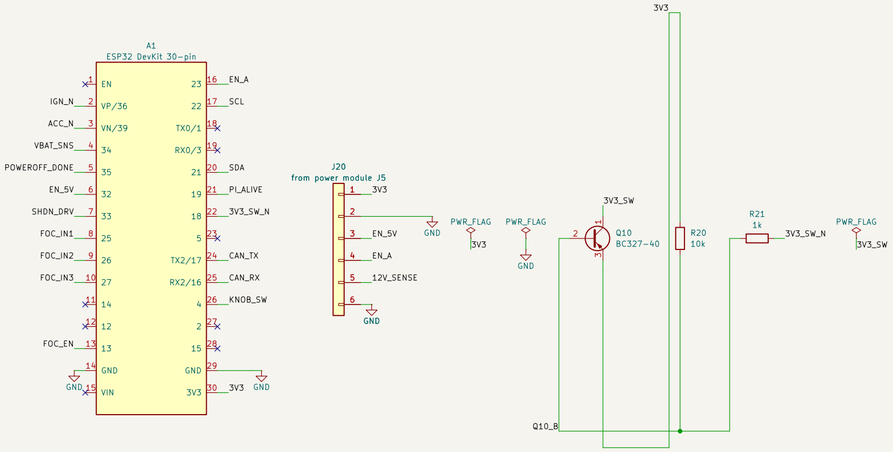
Embedded symbols (lib PERF): generated by scripts/gen-perfboards
Source: docs/design-dash.md (2026-10-29, arch rev 4.3)
Miata NA hardware — TH PCB (owner-fitted TH, JLC SMD)

Sheet: /
File: dash.kicad_sch

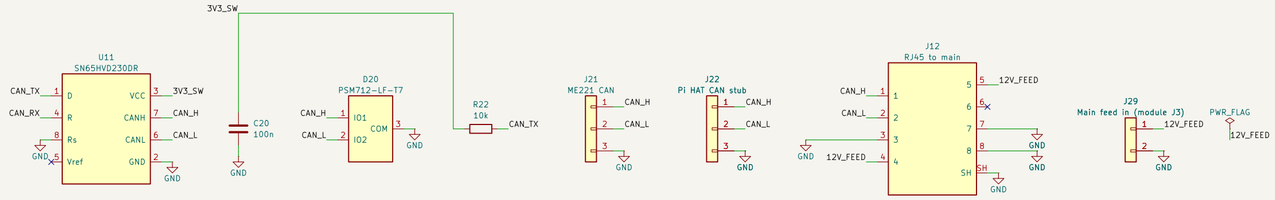
Title: Dash node TH PCB (root)

Size: A3	Date: 2026-10-29	**Rev: 2.0**
KiCad E.D.A. 10.0.7		Id: 1/2

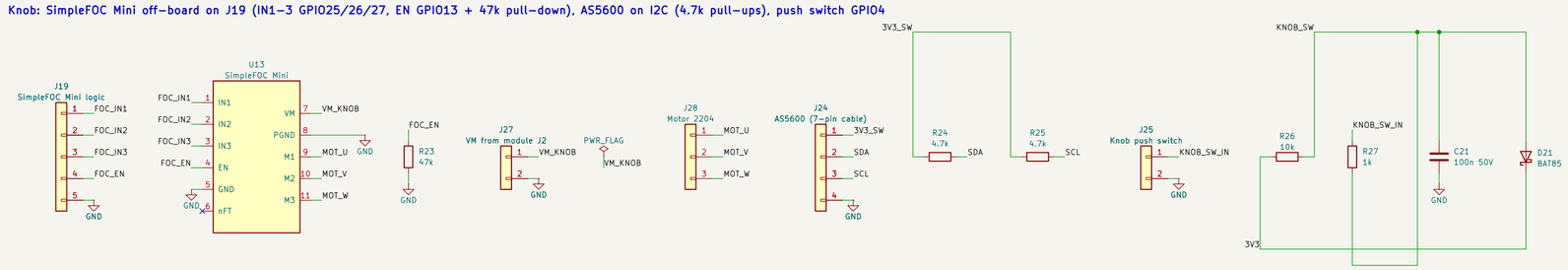
ESP32 30-pin DevKit (fed at its 3V3 pin), power-module cable J20, switched 3V3 for CAN + AS5600 (Q10, GPIO18 low = on)



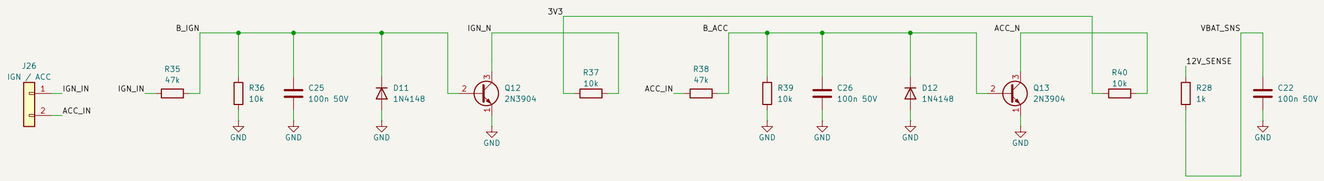
CAN: SN65HVD230DR + 100n + PSM712 (JLC), no termination (mid-bus): ME221 (J21) + RJ45 pins 1/2 to main + Pi HAT stub (J22)



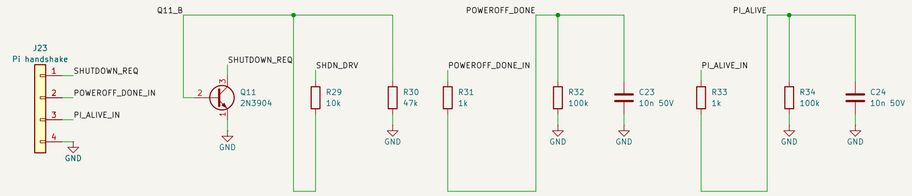
Knob: SimpleFOC Mini off-board on J19 (IN1–3 GPIO25/26/27, EN GPIO13 + 47k pull-down), AS5600 on I2C (4.7k pull-ups), push switch GPIO4



Car inputs IGN -> GPIO36, ACC -> GPIO39 (inverting NPN stage, threshold 3.7 V, f_c 193 Hz); battery sense GPIO34



Pi supervisor: SHUTDOWN_REQ open collector (Q11), POWEROFF_DONE -> GPIO35, PI_ALIVE -> GPIO19 (J23 to the Pi carrier)



Key values (docs/design-dash.md)

Q10: I_B = (3.29 − 0.75 − 0.2)/1k − 0.75/10k = 2.27 mA, load 17.5 mA -> saturated; default OFF (R20)
CAN: SN65HVD230DR Rs = GND (high speed), 500 kbit/s; PSM712 clamps CAN_H/L to −7/+12 V working at J21/J12
Input stage: V_on = 0.65 x (1 + 47/10) = 3.7 V; f_c = 1/(2 pi x 8.25k x 100n) = 193 Hz; −14 V: D clamps, 0.3 mA
Push switch: f_c = 1/(2 pi x 1k x 100n) = 1.59 kHz; I2C t_r = 0.847 x 4.7k x 50 pF = 0.2 us
Q11 SHUTDOWN_REQ: I_B = (2.64 − 0.65)/10k − 0.65/47k = 0.185 mA; sinks 0.33 mA: off in reset (R30)
CAN check (all off): 121 || 120 = 60.2 ohm between J21.1 and J21.2

Embedded symbols (lib PERF): generated by scripts/gen-perfboards
Source: docs/design-dash.md (2026-10-29, arch rev 4.3)
Miata NA hardware — TH PCB (owner-fitted TH, JLC SMD)

Sheet: /Dash node/
File: dash-node.kicad_sch

Title: Dash node TH PCB — ESP32, CAN, knob, Pi supervisor, inputs

Size: A2	Date: 2026-10-29	**Rev: 2.0**
KiCad E.D.A. 10.0.7		Id: 2/2